

Arunabha Majumder

Test Engineer | Laboratory Testing, Troubleshooting & Validation of Mechatronic Systems

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SUMMARY

Mechatronics engineer with hands-on experience in laboratory testing, troubleshooting and validation of systems that combine mechanics, electronics and embedded control. Builds test set-ups, recreates and measures system behaviour, finds the cause of problems and verifies solutions against reference measurements. Structured, curious and used to working across multidisciplinary teams.

KEY RESULTS

- **<10% RMS error:** validated a force-estimation method against a reference force sensor
- **>25% energy reduction** confirmed through laboratory testing of a mechatronic system
- **0.3–0.78% tracking error** after testing and tuning a pneumatic actuator controller

EXPERIENCE

Research Engineer – Mechatronic Systems Testing & Development (PhD Project: VIEXO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Validated a sensorless force-estimation method to below 10% RMS error by testing it against a reference force sensor on a 5-bar planar parallel robot (variable-stiffness actuators and robot Jacobian), then applied it to measure human arm impedance in the gait lab.
- Confirmed a more than 25% energy reduction in laboratory testing by building, instrumenting and testing a hybrid-actuated shoulder exoskeleton (Best Paper Award, IFToMM ISRM 2026).
- Kept prototypes working through repeated test campaigns by troubleshooting and fixing issues across mechanical interfaces, electronics and embedded control: motor, encoder, force sensors and DAQ.
- Designed custom automated test rigs and ran data collection for a 12-participant study across 3 operating modes and 2 load conditions, using EMG and motion-capture equipment.
- Worked with engineers and researchers across mechanical, electronics, control and biomechanics to align on test findings; documented results in peer-reviewed publications.
- Mentored and supervised Bachelor's (B.Tech) students on engineering projects (2023 – 2025).

MTech Researcher – Pneumatic Actuator Testing (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Characterised non-linear actuator behaviour through laboratory testing by building a test rig with valves, pressure sensors and DAQ for in-house pneumatic artificial muscles.
- Reduced position-tracking error to 0.3–0.78% by testing and comparing a neural-network-based controller against a classical PID baseline under varying load.

SKILLS

- **Testing & troubleshooting:** Laboratory testing, test-rig design, troubleshooting, validation against reference measurements, data analysis
- **Electronics & embedded integration:** Sensors, encoders, motors, DAQ, Arduino, LabVIEW (basic), embedded control, pneumatics
- **Lab & measurement equipment:** Force/torque sensors (Forsentec), Qualisys motion capture, EMG, pressure sensors
- **Mechanical & software tools:** SolidWorks (CSWA), technical drawings, FEA, MATLAB/Simulink, Python
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

AWARDS & PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Two peer-reviewed publications: ISRM 2026 (Springer) and IEEE CONECCT 2022

HANDS-ON PROJECTS

- Rehabilitation exoskeleton (Best Major Project Award and TATA Technologies Innovation Grant, 2019), Go-Kart (2017), Efficycle human-electric hybrid vehicle (2018)

REFERENCES

Available upon request.